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Single-carrier frequency-domain turbo equalization without cyclic prefix or zero padding for underwater acoustic communications

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A frequency-domain turbo equalization (FDTE) scheme without cyclic prefix (CP) or zero padding is proposed for single-carrier, multiple-input-multiple-output underwater acoustic communication. In the first iteration of the FDTE, the received continuous data stream is divided into consecutive blocks and a combined inter-block-interference (IBI) cancellation and overlapped windowing scheme is used to diagonalize each data block for low-complexity detection in the frequency domain. Since the second iteration, IBI cancellation and CP reconstruction are applied on each block to enable effective symbol detection. This work extends the authors' previous work on frequency-domain hard-decision equalization to soft-decision turbo equalization so that it not only retains high data transmission efficiency, but also improves the bit error rate performance with slightly increased complexity due to multiple iterations. Its feasibility and effectiveness have been tested by field trial data from the ACOMM09 underwater communication experiment.

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I. INTRODUCTION

High data-rate underwater acoustic (UWA) communications are challenging for two main reasons:^{1–4} First, the available channel bandwidth is very limited; second, the channel condition is very harsh. Specifically, channel variation is rapid and the equivalent discrete-time channel length is often on the order of tens or even hundreds imposing high complexity in the receiver. Besides, Doppler effects, including Doppler shift caused by transceiver relative motion and Doppler spread due to water dynamics, are also significant.⁵ The Doppler-induced non-uniform carrier frequency offset is undesirable and must be mitigated for reliable data detection.

Rapid progress has been made in different aspects for UWA communications recently.^{6–15} The multiple-input-multiple-output (MIMO) technology^{6,9–15} is adopted to fundamentally improve the transmission rate. Frequency-domain detection techniques^{9–11} are utilized to reduce high complexity of equalization due to long multipath delay spread. Turbo detection techniques^{6–9,11–15} are developed to significantly improve the detection performance in terms of bit error rate (BER). Turbo decision-feedback equalizer (DFE) coupled with a digital phase-locked loop has been adopted⁶ to combat phase rotation. Turbo block decision-feedback equalizer¹² is

proposed, which outperforms the conventional DFE with some increased complexity. Turbo linear equalizer (LE)⁸ and an enhanced turbo LE¹⁴ have also been applied to acoustic communications. Frequency-domain turbo equalization^{9,11,15} is also proposed for acoustic communications in the form of orthogonal frequency division multiplexing (OFDM) or single-carrier frequency-domain equalizer (FDE).

Transmission efficiency and detection complexity are conflicting objectives in conventional UWA communications. In general, equalization in the time domain exhibits very high computational complexity for UWA communications due to long channel length. In contrast, the frequency-domain detection has much lower computational complexity, which is a logarithmic function of the channel length. However, frequency-domain detection requires block transmission with the length of guarding cyclic prefix (CP) or zero padding^{9,11,15} being larger than the channel length. CP is used to avoid inter-block interference (IBI) on one hand and to make the channel matrix circular on the other hand. A circular channel matrix, once transformed into the frequency domain, will become a diagonal matrix, thus the detection complexity is kept low. Since CP is a copy of the end part of a block and does not carry extra information bits, it reduces the transmission efficiency, which is critical for UWA communication due to the limited bandwidth.

In this paper, we propose a frequency-domain turbo detection scheme for UWA communications with no need

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to use block transmission. Instead, data are transmitted in continuous stream without guard intervals. On the receiver side, we partition the received continuous stream into consecutive blocks, and then detect each block iteratively. The block size is determined according to the channel coherence time. Since there is no CP available, IBI arises and CP reconstruction is necessary. The IBI in a current block is removed by utilizing the estimated channel and the estimated symbols of a previous block. The CP reconstruction is possible due to the adoption of turbo detection, and has been previously proposed for OFDM systems.¹⁶ The CP in the current iteration is reconstructed with soft symbol estimations from the previous iteration. In the first iteration, no previous symbol estimation is available, then frequency-domain equalization is performed with an overlapped block partition. In the overlapped block partition, the first $N_b = N - L + 1$ symbol vectors are actually detected while the remaining $L - 1$ symbol vectors are treated as interference and are detected in the next block, where L is the channel length and N is the block size. The overlapped block partition enables frequency-domain equalization while suffering performance degradation since part of the symbols in a block are treated as interference of the remaining symbols. The performance degradation in the first iteration, however, will not affect the operation of turbo detection much, since the overall detection performance will increase with the number of iterations.

During data processing, we also notice that phase rotations exist in the equalized symbols. If not properly compensated, phase rotation causes a more severe problem than the loss in signal-to-noise ratio (SNR). A phase estimation and compensation operation is thus applied on the equalized symbols at each iteration. In addition, channel estimation plays a critical role for turbo detection. We combine pilot-aided and decision-directed channel estimation methods to effectively track the time variation of the UWA channel with a low training overhead.

The proposed turbo detection scheme is successfully tested by field trial data measured during the ACOMM09 underwater acoustic communication experiment, which was conducted off the coastline of New Jersey in May 2009. Throughout this paper, $(\cdot)^T$ and $(\cdot)^H$ represent the matrix transpose and Hermitian, respectively.

II. SYSTEM DESCRIPTION

Single-carrier MIMO UWA communications with N_t transducers and N_r hydrophones are considered. To achieve a high data transmission efficiency, the data are organized in the form of frames, with a very large frame size N_f in the order of 10^4 .

The k th received sample within a frame on the m th received hydrophone is represented as

$$y_{m,k} = \sum_{n=1}^{N_t} \sum_{l=0}^{L-1} h_{m,n}(k, l) x_{n,k-l} + v_{m,k}, \quad (1)$$

for $k = 1, 2, \dots, N_f, N_f + 1, \dots, N_f + L - 1$, where $x_{n,k}$ is the k th symbol emitted from the n th transducer, and $v_{m,k}$ is the additive noise with variance σ_v^2 . The subchannel, $\{h_{m,n}(k, l)\}_{l=0}^{L-1}$, between the n th transducer and the m th hydrophone is treated as time-varying due to the long time duration of each frame. All subchannels are assumed to have the same length L , which can be easily implemented by appending zeros at the end of the channel impulse response (CIR) if necessary.

Conventionally, the transmitted symbols, $\{x_{n,k}(1 \leq k \leq N_f)\}_{n=1}^{N_t}$, of the N_t transducers are detected from the received samples, $\{y_{m,k}(1 \leq k \leq N_f + L - 1)\}_{m=1}^{N_r}$, in the time domain. The drawback of the time-domain symbol detection technique is that its complexity increases with the channel length L . To develop a low-complexity detection scheme, we propose to divide the received frame into multiple blocks, and then detect each block independently with frequency-domain equalization technique.

The diagram of the proposed detection scheme is shown in Fig. 1, where a module performing IBI cancellation and CP reconstruction is inserted between the block partition module and the single-carrier, frequency-domain equalizer (SC-FDE) module. The FDE may perform phase rotation estimation and compensation for the equalized symbol. The IBI cancellation and CP reconstruction module is critical for the proposed turbo detection scheme and will be detailed in Sec. III. For turbo detection, the SC-FDE is a soft-input, soft-output module that interacts with the channel decoder in an iterative way via the de-interleaver (Π^{-1}) and the interleaver (Π). The outputs of the SC-FDE, the decoder, and the channel estimator are used for IBI cancellation and CP reconstruction. The FDE output is also feedback to the channel estimator to perform decision-directed channel estimation.^{12,13} In the last iteration of the turbo detection, the decision unit at the last stage of the diagram outputs the hard decisions on transmitted information bits. Details on the proposed turbo detection scheme are provided in the following.

III. PROPOSED LOW-COMPLEXITY TURBO DETECTION

This section presents the details of the low-complexity turbo detection scheme, as shown in Fig. 1, where the received data stream is partitioned into blocks for detection. The choice of the block size is important for balancing the detection

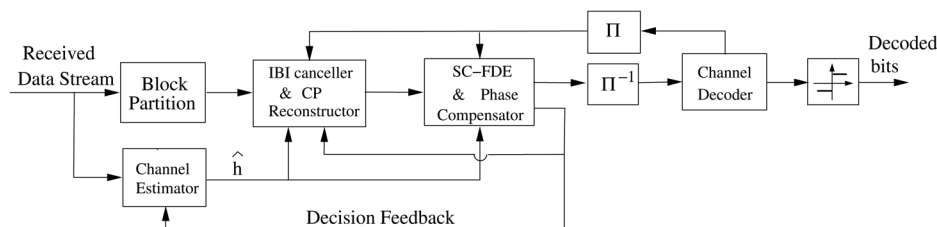


FIG. 1. Block diagram of the proposed turbo detection scheme.

complexity and detection performance. If the block size is too large and its time duration exceeds the channel coherence time, the detection performance is degraded due to the channel variation. On the other hand, if the block size is too small, the overhead with IBI cancellation and CP reconstruction increases and the detection complexity is high. We select the block size based on the channel coherence time.

Block partition by itself, however, cannot guarantee a low-complexity frequency-domain detection, due to the existence of IBI and the missing of CP reconstruction. The post-cursor interference of a previous block can be removed from a current block with the previously estimated symbols and the current channel knowledge. CP reconstruction is made possible in a turbo detection scheme, where the estimated symbols from a previous iteration can be used for the reconstruction. In the first iteration, there is no previous estimation available. In this case, we propose to partition the stream into overlapped blocks, so that the low-complexity detection can still be performed even without CP reconstruction, at the cost of slightly degraded performance. The performance loss, however, does not affect the operation of the turbo detection.

In Secs. III A and III B we discuss the IBI cancellation and CP reconstruction. To facilitate the discussion, we express Eq. (1) in vector form as

$$\mathbf{y}_k = \sum_{l=0}^{L-1} \mathbf{h}(k, l) \mathbf{x}_{k-l} + \mathbf{v}_k, \quad (2)$$

where $\mathbf{y}_k = [y_{1,k}, y_{2,k}, \dots, y_{N_r,k}]^T$, $\mathbf{x}_{k-l} = [x_{1,k-l}, x_{2,k-l}, \dots, x_{N_t,k-l}]^T$, $\mathbf{v}_k = [v_{1,k}, v_{2,k}, \dots, v_{N_r,k}]^T$, and

$$\mathbf{h}(k, l) = \begin{bmatrix} h_{1,1}(k, l) & \dots & h_{1,N_t}(k, l) \\ \vdots & \ddots & \vdots \\ h_{N_r,1}(k, l) & \dots & h_{N_r,N_t}(k, l) \end{bmatrix}. \quad (3)$$

$$\underbrace{\begin{bmatrix} \hat{\mathbf{y}}_1^i \\ \hat{\mathbf{y}}_2^i \\ \vdots \\ \hat{\mathbf{y}}_N^i \end{bmatrix}}_{\hat{\mathbf{y}}^i} = \underbrace{\begin{bmatrix} \mathbf{h}(0) & 0 & \dots & \mathbf{h}(L-1) & \dots & \dots & \mathbf{h}(1) \\ \mathbf{h}(1) & \mathbf{h}(0) & 0 & \dots & \mathbf{h}(L-1) & \dots & \mathbf{h}(2) \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ \mathbf{h}(L-1) & \dots & \mathbf{h}(0) & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & \mathbf{h}(L-1) & \dots & \mathbf{h}(1) & \mathbf{h}(0) \end{bmatrix}}_{\mathbf{h}_{\text{cir}}^i} \underbrace{\begin{bmatrix} \mathbf{x}_1^i \\ \mathbf{x}_2^i \\ \vdots \\ \mathbf{x}_{N_b}^i \\ \mathbf{0}_{N_r(L-1) \times 1} \end{bmatrix}}_{\tilde{\mathbf{x}}^i} + \mathbf{h}^i \underbrace{\begin{bmatrix} \mathbf{0}_{N_r N_b \times 1} \\ \mathbf{x}_{N_b+1}^i \\ \mathbf{x}_{N_b+2}^i \\ \vdots \\ \mathbf{x}_N^i \end{bmatrix}}_{\tilde{\mathbf{x}}^i} + \hat{\mathbf{v}}^i, \quad (9)$$

where the block index i has been dropped from the elements in $\mathbf{h}_{\text{cir}}^i$ for conciseness purposes. From Eq. (9), when $\mathbf{h}^i \tilde{\mathbf{x}}^i$ is treated as an interference term, we are able to detect $\tilde{\mathbf{x}}^i$ via low-complexity frequency-domain equalization since $\mathbf{h}_{\text{cir}}^i$ is a block circular channel matrix.

Based on Eq. (9), $\tilde{\mathbf{x}}^i$ is detected. In other words, the first $N_b = N - L + 1$ symbol vectors of $\mathbf{x}^i$ are actually detected. The remaining $L - 1$ symbol vectors $\mathbf{x}_{N_b+1}^i, \dots, \mathbf{x}_N^i$ have been treated as interference component. Therefore, low-complexity detection without CP reconstruction at the first iteration is

A. Equalization in the first iteration

For the partitioned block with index i , define

$$\mathbf{x}^i = [(\mathbf{x}_1^i)^T, (\mathbf{x}_2^i)^T, \dots, (\mathbf{x}_N^i)^T]^T, \quad (4)$$

$$\hat{\mathbf{y}}^i = [(\hat{\mathbf{y}}_1^i)^T, (\hat{\mathbf{y}}_2^i)^T, \dots, (\hat{\mathbf{y}}_N^i)^T]^T, \quad (5)$$

$$\hat{\mathbf{v}}^i = [(\hat{\mathbf{v}}_1^i)^T, (\hat{\mathbf{v}}_2^i)^T, \dots, (\hat{\mathbf{v}}_N^i)^T]^T, \quad (6)$$

where it is temporarily assumed that the IBI-canceled signal $\hat{\mathbf{y}}^i$ is available (and the $\hat{\mathbf{v}}^i$ is the noise vector after IBI cancellation). We introduce the method that enables the frequency-domain equalization without CP reconstruction. In this case, $\hat{\mathbf{y}}^i$ can be expressed as

$$\hat{\mathbf{y}}^i = \mathbf{h}^i \mathbf{x}^i + \hat{\mathbf{v}}^i, \quad (7)$$

where $\mathbf{h}^i$ is the channel matrix corresponding to the i th block. Since the time duration of a block is short, the channel matrix is treated as time invariant over a block. The channel matrix, $\mathbf{h}^i$, possesses a block Toeplitz structure with the first column (each element is an $N_r \times N_t$ matrix) given as follows:

$$\mathbf{h}_{:,1}^i = [(\mathbf{h}^i(0))^T, \dots, (\mathbf{h}^i(L-1))^T, \mathbf{0}^T, \dots, \mathbf{0}^T]^T, \quad (8)$$

where the time index k has been dropped and each zero matrix has a size of $N_r \times N_t$.

Without CP reconstruction, the channel matrix $\mathbf{h}^i$ is not circular, and thus cannot be diagonalized via discrete Fourier transform (DFT). To achieve a low-complexity detection, we decompose $\mathbf{h}^i \mathbf{x}^i$ in Eq. (7) into two parts: $\mathbf{h}_{\text{cir}}^i \tilde{\mathbf{x}}^i$ and $\mathbf{h}^i \tilde{\mathbf{x}}^i$, and rewrite Eq. (7) as follows:

achieved at the cost of detection efficiency and also some performance degradation. The block partition in the first iteration is demonstrated in Fig. 2, where block i overlaps with block $i - 1$ for $L - 1$ symbol vectors. From Fig. 2, the IBI cancellation is performed as follows:

$$\hat{\mathbf{y}}_k^i = \begin{cases} \mathbf{y}_k^i - \sum_{l=k}^{L-1} \hat{\mathbf{h}}^i(l) \tilde{\mathbf{x}}_{N_b-L+1+l}^{i-1}, & k = 1, \dots, L-1, \\ \mathbf{y}_k^i, & k = L, \dots, N, \end{cases} \quad (10)$$

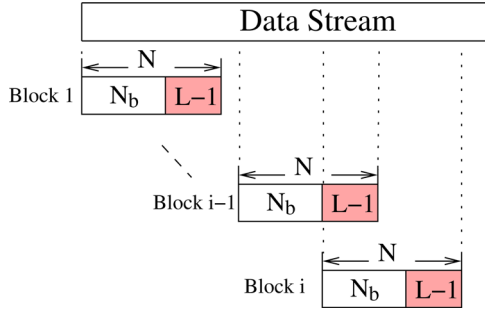


FIG. 2. (Color online) Block partition in the first iteration.

where $\hat{\mathbf{h}}^i(l)$ is the estimated channel for the i th block, and $\bar{\mathbf{x}}_k^{i-1}$ is the k th hard-decision symbol vector of the previous block with index $i-1$.

To perform the low-complexity detection, Eq. (9) is transformed into the frequency domain by applying a size- NN_r normalized block DFT matrix,¹⁵ $\mathbf{F}_{NN_r}$, on both sides. The frequency-domain model is given as

$$\hat{\mathbf{Y}}^i = \mathbf{H}^i \tilde{\mathbf{X}}^i + \mathbf{W}^i, \quad (11)$$

where $\hat{\mathbf{Y}}^i = \mathbf{F}_{NN_r} \hat{\mathbf{y}}^i$, $\tilde{\mathbf{X}}^i = \mathbf{F}_{NN_r} \tilde{\mathbf{x}}^i$ with $\mathbf{F}_{NN_r}$ being a size- NN_r normalized block DFT matrix,¹⁵ $\mathbf{W}^i = \mathbf{F}_{NN_r} \mathbf{w}^i$ with $\mathbf{w}^i = \mathbf{h}^i \tilde{\mathbf{x}}^i + \mathbf{v}^i$, and $\mathbf{H}^i = \mathbf{F}_{NN_r} \mathbf{h}_{\text{cir}}^i \mathbf{F}_{NN_r}^H$. The frequency-domain symbol vector $\tilde{\mathbf{x}}^i$ is estimated, then transformed back into its time-domain counterpart inside the soft-input, soft-output FDE.¹⁵ Due to the Doppler spread in UWA communications, the equalized symbols suffer phase rotations,¹³ so we perform phase estimation and compensation after the equalization, as will be shown in Sec. IV. Significant performance gain can be achieved with the phase estimation and compensation operation. The soft output of the SC-FDE is computed with the phase-compensated symbol estimations.^{15,17} The channel decoder takes the soft output of the SC-FDE as input and feeds back new soft information to the SC-FDE for the next iteration.

B. Equalization in the second iteration and beyond

For the second iteration and beyond, the soft output in a previous iteration from the channel decoder is available for the SC-FDE. The soft feedback is usually in the form of log-likelihood ratio, with which the *a priori* probability, $P(x_{n,k}^i = \alpha)$, of a symbol, $x_{n,k}^i$, can be computed.¹⁷ The value α is taken from the constellation set $\mathcal{S}$. With $P(x_{n,k}^i = \alpha)$, the *a priori* symbol estimation is computed as follows:

$$\bar{x}_{n,k}^i = \sum_{\alpha \in \mathcal{S}} \alpha \cdot P(x_{n,k}^i = \alpha). \quad (12)$$

The soft symbol estimations can be used for CP reconstruction. Temporarily assuming that the IBI-canceled received signal $\hat{\mathbf{y}}^i$ is available, the CP reconstruction is performed as follows:

$$\tilde{\mathbf{y}}_k^i = \begin{cases} \hat{\mathbf{y}}_k^i + \sum_{l=k}^{L-1} \hat{\mathbf{h}}^i(l) \bar{\mathbf{x}}_{N+1-l}^i, & k = 1, 2, \dots, L-1, \\ \hat{\mathbf{y}}_k^i, & k = L, \dots, N, \end{cases} \quad (13)$$

where $\bar{\mathbf{x}}_k^i = [\bar{x}_{1,k}^i, \bar{x}_{2,k}^i, \dots, \bar{x}_{N_r,k}^i]^T$ is the *a priori* soft symbol estimation vector. After CP reconstruction, the vector $\tilde{\mathbf{y}}^i$ is given as

$$\tilde{\mathbf{y}}^i \approx \mathbf{h}_{\text{cir}}^i \mathbf{x}^i + \mathbf{v}^i, \quad (14)$$

where approximation is used since there exists estimation error in $\bar{\mathbf{x}}_k^i$. The model in Eq. (14) is ready for low-complexity detection. It is noted that the accuracy of $\bar{\mathbf{x}}_k^i$ increases with the turbo iterations, so the detection with Eq. (14) approaches that for a block transmission with CP.

The IBI cancellation for the second iteration and beyond is similar to the first iteration, as shown in Eq. (10), except that the N_b in the subscript of $\bar{\mathbf{x}}^{i-1}$ shall be changed to N . In other words, there is no need for overlapped block partition and thus no loss in the detection efficiency since the second iteration, attributed to the CP reconstruction.

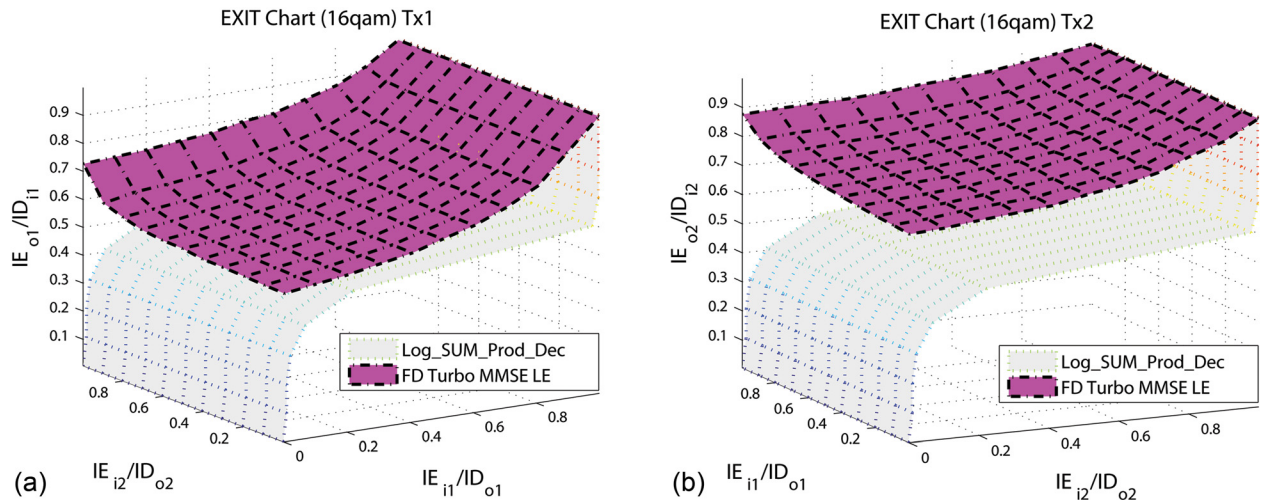


FIG. 3. (Color online) EXIT chart. (a) Transducer 1. (b) Transducer 2.

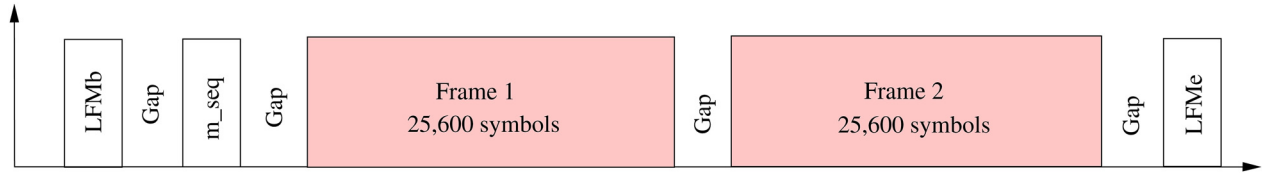


FIG. 4. (Color online) Packet structure for ACOMM09.

C. Channel decoding

A binary low-density parity-check (LDPC) channel code with coding rate $1/2$ has been used in the ACOMM09 experiment. The code word length can be 1024, 2048, or 4096. The decoding is performed with the log-domain Sum-Product Algorithm.¹⁸ Details are omitted here for brevity.

D. Convergence analysis

The EXIT chart¹⁹ is used to explain the convergence of the turbo detection with the exchange of soft information between the equalizer and decoder. Figures 3(a) and 3(b) depict the EXIT chart for the 2 by 2 MIMO system with the estimated frequency selective channels from ACOMM09 experiment and 16QAM modulation at SNR = 18 dB, where two three-dimensional EXIT charts were generated with IE_{i1} and IE_{i2} as the input and IE_{o1} or IE_{o2} as the output. This is because two data streams were transmitted in the MIMO channel, the *a priori* information from both transmitted streams may affect both outputs of the equalizer simultaneously. Besides, the soft information of the decoder is related with those of the equalizer as $ID_{o,n} = IE_{i,n}$ and $ID_{i,n} = IE_{o,n}$ with $n = 1, 2$. The extrinsic information transfer characteristics of the equalizer (SC-FDE) is demonstrated by the upper structure, and the lower surface shows the swapped transfer characteristics between the output extrinsic information and the input *a priori* information of the LDPC decoder. There is clearly a zigzag path between these two surfaces for both EXIT charts. This verified that the proposed frequency-domain turbo equalization has good convergence with the underwater channel.

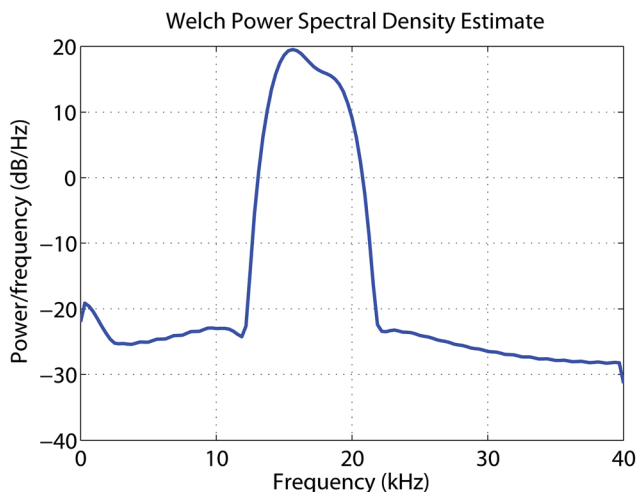


FIG. 5. (Color online) Power spectrum of a received packet.

IV. FIELD TEST RESULTS FROM THE ACOMM09 EXPERIMENT

A single-carrier underwater communication experiment named ACOMM09 was conducted off the coastline of New Jersey in May 2009. The transmitter is a two-transducer array. On the transmitter side, bit-interleaved coded modulation was used in the time domain and multiplexing was used in the space domain. A random interleaver was adopted, and the channel encoder is a rate $1/2$ LDPC code. Different modulations of Quadrature Phase-Shift Keying (QPSK), 8 Phase-Shift Keying (8PSK), and 16 Quadrature Amplitude Modulation (16QAM) were used. The transmission rate for a single transducer was 5 kilo symbols per second (ksps), and the carrier frequency was $f_c = 17$ kHz. Two set of receivers, named ACDS2 and ACDS3, were deployed. Each receiver consisted of eight hydrophones. The ACDS2 and ACDS3 were 2 and 3 km away from the transmitter, respectively. Fixed communications with no transceiver relative motion were performed.

The structure of a transmitted packet in the ACOMM09 experiment, is shown in Fig. 4. From the Fig. 4, two frames are transmitted within one packet. The frame size is $N_f = 25\,600$, which is much larger than the block size used in a block transmission with CP.^{9,11,15} The linear frequency modulation (LFM) sequences LFMb and LFMe are used for synchronization and Doppler shift estimation.¹³

During the experiment, the received packets were recorded in files for offline detection. In Fig. 5, the power spectrum of a recorded packet is plotted. A passband filtering has been applied on the time-domain signal so the spectrum

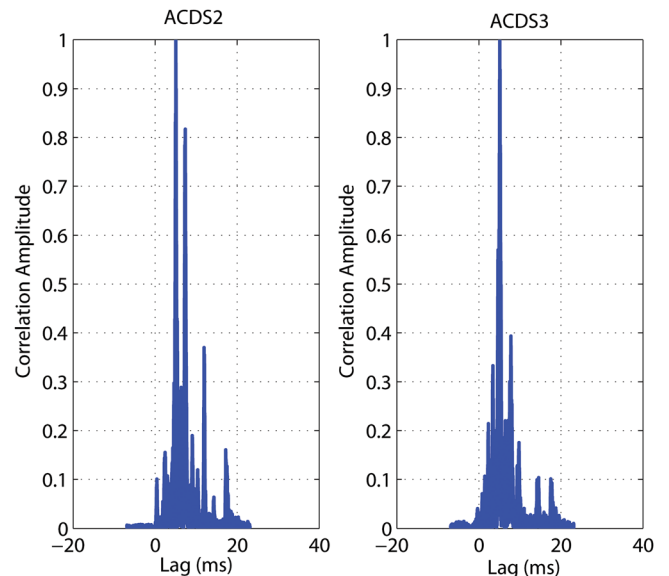


FIG. 6. (Color online) Examples of normalized LFMb correlations.

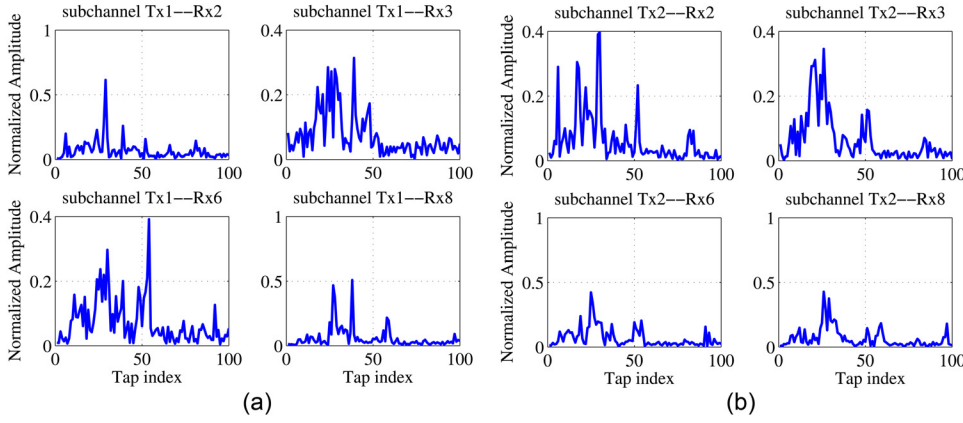


FIG. 7. (Color online) Estimated MIMO channel (ACDS2). (a) Transducer 1. (b) Transducer 2.

curve is smooth. It is clear that the signal frequency component centers at the carrier frequency $f_c = 17$ kHz, and spans a range larger than 5 kHz due to the excess bandwidth of the pulse shaping filter.

In Fig. 6, examples of the normalized LFM correlations for the ACDS2 and ACDS3 are respectively shown, where the correlation peaks indicate the coarse synchronization points. In each panel, the span of the LFM correlation is about 20 ms, which corresponds to a channel length of $L \approx 20 \text{ ms}/T_s = 100$ in terms of the symbol period. The symbol period T_s is equal to 0.2 ms since the symbol rate per transducer is 5 kbps. The correlation peak is in the middle of the correlation span, so the CIRs are non-minimum phase.

The estimated MIMO channel for ACDS2 is shown in Fig. 7, where the energy of each subchannel has been normalized to one. The notations “Tx” and “Rx” denote a transducer and a hydrophone, respectively. Obviously, the subchannels are sparse and non-minimum phase, which is consistent with the observations in Fig. 5. In Fig. 8, the sequence of estimated channels of different blocks within a frame is demonstrated. For comparison purposes, the chan-

nel energy is not normalized. It is obvious that the channel varies over one frame.

Phase rotation caused by Doppler spread exists in the equalized symbols for UWA communications, especially when the block size is large.¹³ A phase estimation and correction method for PSK modulation has already been demonstrated in previous work.¹³ In this paper, we propose a phase estimation and compensation method for 16QAM modulation. For an equalized block with index i , the mean power of the estimated symbols corresponding to the n th transducer is computed as follows:

$$\varepsilon_n = \frac{1}{N_b} \sum_{k=1}^{N_b} |\hat{x}_{n,k}^i|^2. \quad (15)$$

Then, those equalized symbols with a modulus in the range $[1.1708\sqrt{\varepsilon_n}, 1.5125\sqrt{\varepsilon_n}]$ are selected for phase rotation estimation. The reason for selecting estimated symbols with a large power is that they are less sensitive to additive noise. The idea for phase rotation estimation is to find the average difference between the phases of those selected symbol estimations and the phases of their hard-decision counterparts. The estimated phase rotation is then used for compensating all estimated symbols within a block. It is noted that the phase rotation estimation and compensation is performed for each transducer independently, since symbols from different transducers may undergo different phase corruption. In Fig. 9, an example of phase rotation estimation and compensation is shown. In the left panel, the phase rotation in the equalized symbols is clearly shown. The selected symbols for phase rotation estimation fall into the region between the two circles, and the four big dots (A, B, C, D) represent the hard decisions for those selected symbols. The phase-corrected symbol estimations are shown in the right panel.

The detection gain obtained from the phase compensation (PC) is often significant. In Table I, detection results with and without phase compensation are listed for both ACDS2 and ACDS3. The result with 16QAM modulation is shown. Five packets are detected for ACDS2, and four packets are detected for ACDS3. The N_{iter} in Table I denotes the number of iterations for the turbo equalization. From

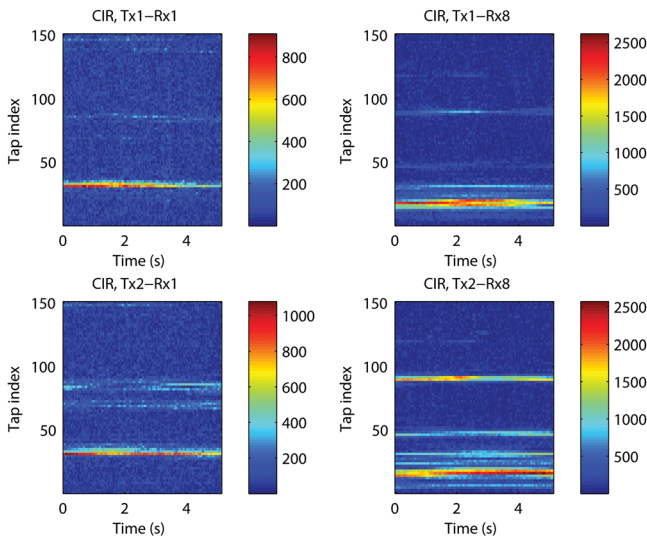


FIG. 8. (Color online) Demonstration of channel time variation within one frame.

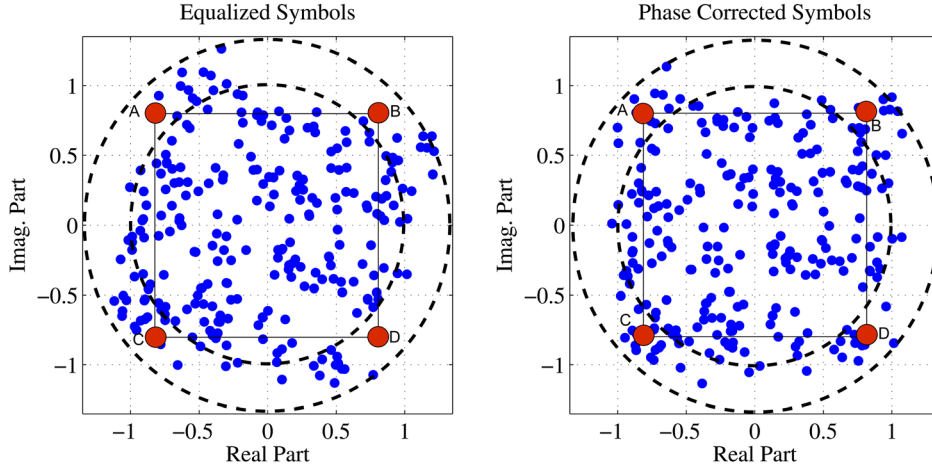


FIG. 9. (Color online) Demonstration of phase rotate estimation and compensation with 16QAM modulation (the four big dots in both panels are the four nominal 16QAM points within that circular band).

Table I, using PC consistently improves the performance. For packet 1 of either receiver, the error numbers are dramatically reduced and the performance increases with iterations by using PC. Without using PC, the detection completely failed.

Before the experimental results are further presented, we show the turbo detection process by plotting the estimated symbols at the output of the soft-input, soft-output FDE and the LDPC channel decoder over multiple iterations. In Fig. 10(a), the 8PSK symbol estimations are demonstrated for ACDS2. Obviously, the symbol detection at the output of the decoder is nearly perfect in the first iteration. In the second iteration, the output of the soft-input, soft-output FDE has no errors. The results for ACDS3 are shown in Fig. 10(b), where error-free detection is obtained in the third iteration. Both results have shown the powerful detection capa-

bility of turbo equalization. The results with 16QAM modulation are also included in Figs. 11(a) and 11(b) for the two receivers. The observations are similar to the 8PSK case.

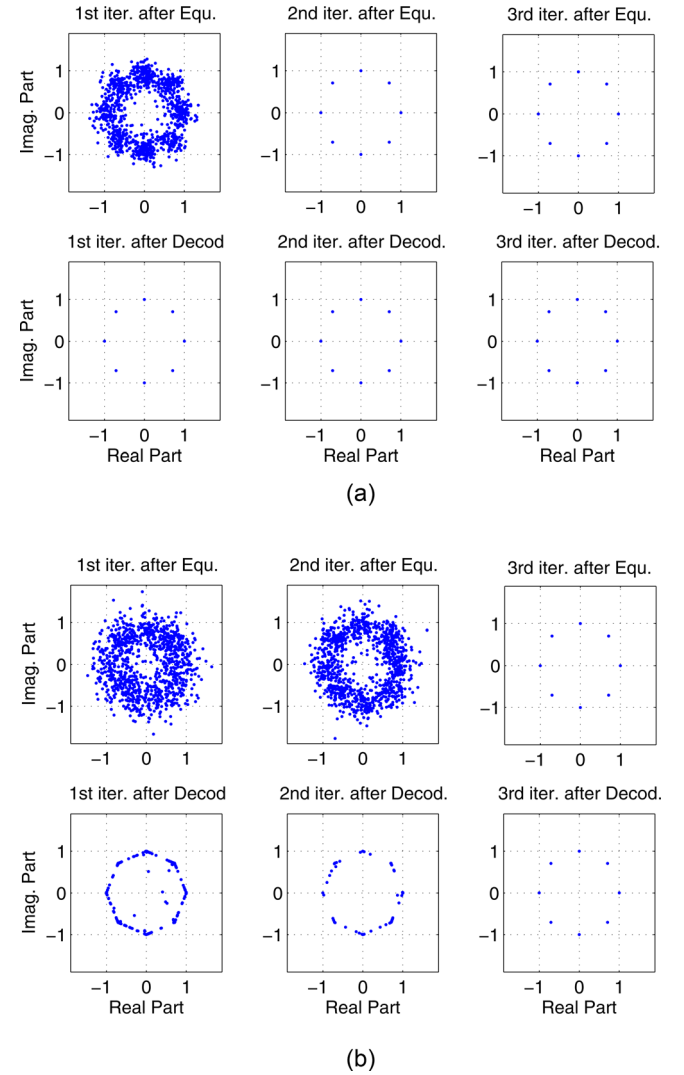


FIG. 10. (Color online) Demonstration of turbo equalization process with phase compensation for 8PSK modulation. (a) ACDS2. (b) ACDS3.

TABLE I. Number of error bits for 2×8 MIMO with 16QAM modulation (184 320 information bits per packet). Cases (a) and (b) are results with and without phase compensation, respectively.

		N_{iter}			
Pocket		1	2	3	4
ACDS2	1(a)	421	97	27	10
	1(b)	64 425	70 470	73 186	74 641
	2(a)	365	86	26	13
	2(b)	2 574	1 754	897	874
	3(a)	314	59	9	4
	3(b)	1 578	2 547	3 658	5 987
	4(a)	197	47	7	2
	4(b)	275	124	21	14
	5(a)	0	0	0	0
	5(b)	356	117	57	31
ACDS3	1(a)	645	87	37	10
	1(b)	58 741	69 541	71 548	75 412
	2(a)	541	76	31	13
	2(b)	4 412	5 472	5 874	6 587
	3(a)	361	55	27	12
	3(b)	1 578	2 547	3 658	5 987
	4(a)	50	0	0	0
	4(b)	55	1	1	1

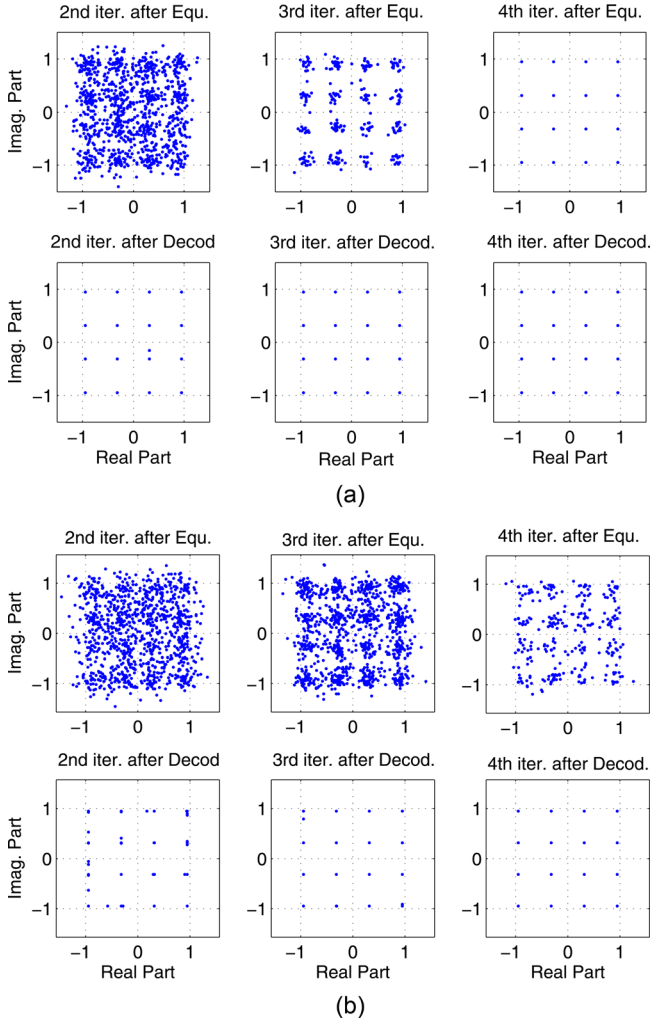


FIG. 11. (Color online) Demonstration of turbo equalization process with phase compensation for 16QAM modulation. (a) ACDS2. (b) ACDS3.

For each frame, pilot blocks of size 512 are inserted periodically for every 5120 symbols. In total, there are five pilot blocks within each frame, resulting in a pilot overhead of 10%. The pilot block provides an initial channel estimation for detecting the first partitioned block following it, and decision-directed channel estimation is used for detecting remaining partitioned blocks. During the packet detection, the number of LDPC decoding iterations has been chosen as 15.

TABLE II. Detection results (BERs) for 2×8 MIMO, $N_b = 128$.

		N_{iter}			
Modulation		1	2	3	4
ACDS2	QPSK	6.2×10^{-5}	0	0	0
	8PSK	3.6×10^{-4}	1.8×10^{-4}	2.6×10^{-5}	0
	16QAM	1.8×10^{-3}	3.2×10^{-4}	1.4×10^{-5}	5.6×10^{-5}
ACDS3	QPSK	1.0×10^{-4}	0	0	0
	8PSK	4.4×10^{-4}	2.1×10^{-4}	4.4×10^{-5}	2.3×10^{-6}
	16QAM	3.0×10^{-3}	4.4×10^{-4}	2.3×10^{-5}	8.9×10^{-5}

TABLE III. Detection results (BERs) for 2×8 MIMO, $N_b = 256$.

		N_{iter}			
Modulation		1	2	3	4
ACDS2	QPSK	1.6×10^{-4}	0	0	0
	8PSK	1.8×10^{-3}	3.1×10^{-4}	4.9×10^{-5}	0
	16QAM	3.2×10^{-3}	7.1×10^{-4}	1.7×10^{-4}	7.1×10^{-5}
ACDS3	QPSK	2.2×10^{-4}	4.4×10^{-6}	0	0
	8PSK	1.8×10^{-3}	3.9×10^{-4}	8.1×10^{-5}	2.3×10^{-6}
	16QAM	4.9×10^{-3}	6.7×10^{-4}	2.9×10^{-4}	1.1×10^{-4}

For ACDS2, 11 packets with QPSK modulation, 8 packets with 8PSK modulation, and 5 packets with 16QAM modulation are processed. For ACDS3, the number of detected packets are 11, 7, and 4, corresponding to the three modulations. By resorting to the channel scattering function estimated with the m -sequence,¹³ we find the channel coherence time as $T_c \approx 125$ ms, corresponding to a Doppler spread over the range $[-4, 4]$ Hz. Based on the channel coherence time, three different partition block sizes: $N = 227$, $N = 355$, and $N = 611$ (or $N_b = 128$, $N_b = 256$, and $N_b = 512$), are chosen. The corresponding block time durations are 45.4, 71, and 122.2 ms. The BER is used as a performance metric. The results with the three different partition block sizes are listed in Tables II–IV.

From Tables II–IV, error-free detection can be obtained for QPSK modulation despite the value of N_b , for both ACDS2 and ACDS3. Two or three iterations are sufficient for error-free detection. With 8PSK modulation, the ACDS2 achieves error-free detection when $N_b = 128$ and $N_b = 256$, and has a BER level of 10^{-4} with $N_b = 512$. The performance of ACDS3 is not as good as ACDS2, and error-free detection is not reached regardless of the partitioned block size. With N_b equal to 128 or 256, the BER level is 10^{-6} . With N_b equal to 512, the BER level is 10^{-4} . For the 16QAM modulation, error-free detection is not achieved in all cases. The BER level ranges from 10^{-3} to 10^{-5} for ACDS2 and ACDS3. In summary, a smaller partitioning block leads to a better BER performance as expected. The overall performance of ACDS2 with a transmission range of 2 km is better than that of ACDS3 with a transmission range of 3 km, due to its higher SNR of the received signal.

TABLE IV. Detection results (BERs) for 2×8 MIMO, $N_b = 512$.

		N_{iter}			
Modulation		1	2	3	4
ACDS2	QPSK	2.8×10^{-4}	6.7×10^{-6}	0	0
	8PSK	1.1×10^{-2}	1.4×10^{-3}	2.5×10^{-4}	2.5×10^{-4}
	16QAM	2.5×10^{-2}	6.0×10^{-3}	6.3×10^{-4}	6.1×10^{-4}
ACDS3	QPSK	3.2×10^{-4}	1.8×10^{-5}	0	0
	8PSK	1.5×10^{-2}	1.8×10^{-3}	3.8×10^{-4}	3.8×10^{-4}
	16QAM	3.4×10^{-2}	8.1×10^{-3}	9.9×10^{-4}	9.8×10^{-4}

V. CONCLUSION

Low-complexity turbo detection for single-carrier MIMO UWA communication was proposed. Compared with conventional low-complexity frequency-domain detection, operation of the proposed scheme did not require CP-based block transmission, thus a high data transmission efficiency was achieved. The proposed scheme partitioned the received continuous stream into small blocks, and detected each block in the frequency domain by relying on IBI cancellation and CP reconstruction. To mitigate the phase rotation in the equalized symbols, a phase rotation estimation and compensation method was introduced and significantly improved the detection performance. The proposed low-complexity scheme was tested by field trial data measured in the ACOMM09 undersea communication experiment. Detection results showed a consistently good detection performance of the proposed scheme.

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